

Emotion Detection from Text Using NLP and Machine Learning Approaches

Project Title :
Emotion Detection from Text Using NLP and Machine Learning Approaches

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1. Introduction

The rapid growth of social media and online communication has produced large amounts of text in which users express their feelings, opinions, and reactions. Automatically identifying these emotions is an important Natural Language Processing task that can support applications such as public-opinion analysis, customer-feedback analysis, crisis monitoring, and social-media research.

Emotion detection in Arabic is particularly challenging because online Arabic contains Modern Standard Arabic, regional dialects, spelling variations, repeated characters, emojis, hashtags, English words, and informal writing styles. Negation, short context, and emotional overlap can also make the intended emotion difficult to identify.

This project aims to develop a supervised multiclass NLP system that automatically classifies Arabic text into different emotion categories. It uses the ArPanEmo Arabic Pandemic Emotion Dataset, which contains Arabic posts collected from several online platforms and manually assigned emotion labels.

Two main NLP approaches are investigated. The first is a traditional machine-learning approach based on word- and character-level TF-IDF features with Logistic Regression. The second uses a fine-tuned AraBERT transformer model to learn contextual representations of Arabic text. The approaches are evaluated and compared using standard classification metrics, confusion matrices, and error analysis.

The following sections describe the dataset, preprocessing and feature-extraction methodology, model implementation, experimental results, identified challenges, and possible future improvements.

2. Dataset Description

This project uses the **ArPanEmo (Arabic Pandemic Emotion Dataset)**, an open-source dataset developed by Maha Jarallah Althobaiti for fine-grained emotion classification in Arabic online content. The dataset contains posts collected between March 2020 and March 2022 from Twitter, YouTube comments, and online newspaper comments.

The posts discuss the COVID-19 pandemic and its effects on healthcare, infection, vaccination, lockdown, remote education, government decisions, the economy, and daily life. Most posts are written in Saudi Arabic, although the dataset also includes Modern Standard Arabic, other dialectal expressions, English words, hashtags, emojis, and informal social-media writing.

The complete dataset contains **11,128 manually labeled Arabic posts**, divided into two official CSV files:

Dataset File	Records	Percentage
`ArPanEmo_train.csv`	8,901	80%
`ArPanEmo_test.csv`	2,227	20%
Total	**11,128**	**100%**

Each file contains three main columns:

Column	Description
`number`	Twitter post ID, or zero for YouTube and newspaper comments
`post`	Original Arabic text of the post
`label`	Manually assigned emotion category

Each post is assigned one of eleven emotion categories:

Anger, Anticipation , Confusion , Disgust , Fear, Happiness, Neutral , Optimism , Pessimism , Sadness ,Surprise

An important limitation is that many Twitter records contain only an ellipsis (`...`) instead of the original text because the tweet content cannot be redistributed. Dataset inspection found **1,683 available texts** and **9,445 unavailable texts**. This limitation significantly reduces the amount of text that can be used for preprocessing, feature extraction, and model training. The handling of these records and the final usable dataset are explained in the Methodology section.

3. Methodology

This project follows a supervised multiclass text-classification methodology. The workflow consists of dataset inspection, Arabic text preprocessing, training and validation data preparation, feature extraction, model implementation, and performance evaluation.

The available Arabic posts were cleaned and normalized while preserving important emotional signals such as negation, emojis, and expressive punctuation. Conflicting labels, duplicate

cleaned texts, and train-test overlap were removed to improve data quality and prevent leakage. The cleaned training data were then divided into training and validation subsets, while a separate cleaned test set was retained for final evaluation.

Two main NLP approaches were implemented. The first combines word- and character-level TF-IDF features with a multiclass Logistic Regression classifier. The second fine-tunes AraBERT using the cleaned Arabic text. A majority-class baseline was also implemented as a simple reference.

The approaches were evaluated using Accuracy, Macro Precision, Macro Recall, Macro F1-score, Weighted F1-score, classification reports, confusion matrices, and error analysis. Detailed preprocessing and feature-extraction procedures are presented in the following sections.

4. Text Preprocessing

The preprocessing stage was applied to prepare the Arabic posts for feature extraction and model training. The objective was to remove unavailable and irrelevant content, reduce unnecessary writing variations, and preserve elements that may provide useful emotional information.

4.1 Removing Unavailable Text

Records were removed when the `post` column contained:

- Empty or whitespace-only values.
- One or more dots such as `...`.
- Numeric identifiers without actual text.
- No Arabic or English letters.

This step removed 7,473 training records and 1,972 test records because these records did not contain usable text.

4.2 Text Cleaning and Normalization

The following operations were applied to each available post:

- HTML entities were decoded and HTML tags were removed.
- URLs were replaced with the token `URL`.
- User mentions were replaced with the token `USER`.
- The hashtag symbol `#` was removed while preserving the hashtag word.
- Arabic diacritics and tatweel were removed.
- Different forms of Alef (`آ`, `إ`, `أ`, and `ٱ`) were normalized to `ا`.
- The letter `ى` was normalized to `ي`.
- Characters repeated three or more times were reduced to two occurrences.
- Unnecessary symbols were removed.
- Repeated spaces and line breaks were normalized.

4.3 Preserving Emotional Information

Negation words were preserved because they can reverse the emotional meaning of a sentence. Emojis and expressive punctuation, such as exclamation and question marks, were also retained because they may indicate the type or intensity of an emotion.

Arabic and English words were preserved to support posts containing mixed-language expressions. Stop-word removal and stemming were not applied because they could remove useful emotional or contextual information.

4.4 Removing Conflicts, Duplicates, and Leakage

After text normalization, cleaned texts assigned to different emotion labels were considered conflicting and removed. Duplicate cleaned texts were also removed to prevent repeated examples from having excessive influence on the models.

The cleaned training and test files were compared, and overlapping texts were removed from the test set to prevent data leakage and ensure that the models were evaluated on unseen examples.

4.5 Preprocessing Results

Processing Result	Training	Test
Original records	8,901	2,227
Unavailable records removed	7,473	1,972
Conflicting-label records removed	6	0
Duplicate cleaned texts removed	3	2
Train-test overlap removed	N/A	3
Final usable records	**1,419**	**250**

The final cleaned training data were divided into 1,135 training records and 284 validation records.

The preprocessing stage generated the following files:

```
ArPanEmo_train_clean.csv
ArPanEmo_test_clean.csv
ArPanEmo_train_split.csv
ArPanEmo_validation_split.csv
```

Each file includes the original text in the `post` column and the processed version used by the models in the `clean_text` column.

5. Feature Extraction

Machine-learning models cannot process raw Arabic text directly. Therefore, the cleaned posts were converted into numerical features using **Term Frequency-Inverse Document Frequency (TF-IDF)**.

TF-IDF assigns higher weights to terms that are important in a specific post and lower weights to terms that occur frequently across many posts. The feature-extraction stage was applied to the `clean_text` column produced during preprocessing.

Two types of TF-IDF features were extracted and combined:

1. Word-level TF-IDF features.
2. Character-level TF-IDF features.

5.1 Word-Level TF-IDF

Word-level TF-IDF extracts individual words and short word sequences from the cleaned posts.

The following configuration was used:

```
analyzer = word
ngram_range = (1,2)
min_df = 2
max_features = 60,000
sublinear_tf = True
```

The `(1,2)` n-gram range extracts unigrams and bigrams. Bigrams are useful for preserving short expressions involving negation, such as `مش سعيد`, whose meaning differs from the word `سعيد` alone.

The fitted word vectorizer produced **3,037 word-level features**.

5.2 Character-Level TF-IDF

Character-level TF-IDF extracts sequences of characters from inside word boundaries.

The following configuration was used:

```
analyzer = char_wb
ngram_range = (3,5)
min_df = 2
max_features = 80,000
sublinear_tf = True
```

Character n-grams help the model handle:

- Arabic dialectal expressions.
- Spelling variations.
- Prefixes and suffixes.
- Typing mistakes.
- Informal repeated characters.
- Words not previously seen in exactly the same form.

The fitted character vectorizer produced **17,504 character-level features**.

5.3 Combining the Features

The word and character TF-IDF representations were combined using Scikit-learn's `FeatureUnion`.

```
Cleaned Arabic Text
        ↓
Word TF-IDF + Character TF-IDF
        ↓
Combined Sparse Feature Matrix
        ↓
Logistic Regression Classifier
```

The final feature representation contained:

Feature Type	Number of Features	
Word-level TF-IDF	3,037	
Character-level TF-IDF	17,504	

Feature Type	Number of Features	
Combined Features	**20,541**	

The resulting training feature matrix had the shape:

```
(1,135, 20,541)
```

This represents 1,135 training posts using 20,541 numerical features. The matrix was stored in a sparse format because most posts contain only a small percentage of all possible features.

5.4 Preventing Feature-Extraction Leakage

The TF-IDF extractor was fitted only on `ArPanEmo_train_split.csv`. The validation and test data were transformed using the same fitted extractor:

```
Fit TF-IDF on Training Data
        ↓
Transform Training Data
        ↓
Transform Validation Data
        ↓
Transform Test Data
```

The validation and test texts were not used to construct the TF-IDF vocabulary. This prevents information from the evaluation data from leaking into model training.

The extracted TF-IDF features were used only by the Logistic Regression model. AraBERT received the cleaned text directly through its tokenizer and did not use the TF-IDF feature matrices.

6. Model Development

Three classification models were implemented: a majority-class baseline, TF-IDF with Logistic Regression, and fine-tuned AraBERT. The baseline was used only as a reference, while Logistic Regression and AraBERT represent the two main NLP approaches required by the project.

6.1 Majority-Class Baseline

A majority-class baseline was implemented using Scikit-learn's `DummyClassifier` with:

```
strategy = most_frequent
```

The baseline identifies the most frequent emotion in the training subset and predicts that emotion for every input text. In the cleaned training data, the most frequent class was Happiness.

The baseline does not analyze the text or learn linguistic patterns. Its purpose is to establish a minimum reference performance that the two main models should exceed.

6.2 TF-IDF with Logistic Regression

The first main approach uses a multiclass Logistic Regression classifier trained on the combined word- and character-level TF-IDF features.

The model was configured as follows:

```
Classifier = Logistic Regression
Maximum iterations = 1,000
```

```
Random state = 42
```

The development process was:

```
Cleaned Training Text
        ↓
Word and Character TF-IDF
        ↓
Combined Sparse Feature Matrix
        ↓
Logistic Regression Training
        ↓
Validation and Test Predictions
```

The classifier was trained on the 1,135-record training subset. The fitted TF-IDF extractor was then used to transform the validation and test data without rebuilding the vocabulary.

The trained classifier and feature extractor were saved together in:

```
tfidf_logistic_regression.joblib
```

This allows future Arabic texts to be transformed and classified using the same fitted components.

6.3 Fine-Tuned AraBERT

The second approach uses the pretrained Arabic transformer:

```
aubmindlab/bert-base-arabertv02
```

The model and its matching tokenizer were loaded using the Hugging Face Transformers library. Emotion labels were converted into numerical values using `LabelEncoder`.

Each cleaned post was tokenized and prepared using:

```
Maximum sequence length = 128 tokens
Truncation = True
Padding = True
```

AraBERT was fine-tuned using the following settings:

Parameter	Value
Training epochs	3
Learning rate	2×10−52 \times 10^{-5}
Training batch size	8
Evaluation batch size	8
Weight decay	0.01
Random seed	42
Model-selection metric	Macro F1

The model was evaluated after each epoch, and the checkpoint with the best validation Macro F1-score was selected. Only one checkpoint was retained to control storage usage.

The fine-tuned model and tokenizer were saved in:

```
arabert_best_model/
```

6.4 Model Training and Prediction

Both main approaches were trained using:

```
ArPanEmo_train_split.csv
```

The validation file was used for preliminary evaluation and model comparison:

```
ArPanEmo_validation_split.csv
```

The final predictions were produced using the held-out test file:

```
ArPanEmo_test_clean.csv
```

The test data were not used to train either model. Using the same training, validation, and test partitions allowed a fair comparison between Logistic Regression and AraBERT.

6.5 Model Outputs

For each main model, the pipeline generated:

- A validation classification report.
- A test classification report.
- A validation confusion matrix.
- A test confusion matrix.
- Validation misclassification records.
- Test misclassification records.

All final model settings and evaluation metrics were also stored in:

```
phase2_complete_summary.json
```

7. Experimental Results

The majority-class baseline, TF-IDF with Logistic Regression, and fine-tuned AraBERT were evaluated using Accuracy, Macro Precision, Macro Recall, Macro F1-score, and Weighted F1-score. Macro F1 was considered particularly important because it gives equal importance to each represented emotion class despite differences in class size.

7.1 Validation Results

The following table presents the results on the 284-record validation set:

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Majority Baseline	13.03%	1.30%	10.00%	2.31%	3.00%
TF-IDF + Logistic Regression	**50.00%**	**51.45%**	**47.52%**	**47.42%**	**48.69%**
AraBERT	46.13%	46.86%	43.14%	42.16%	44.12%

Both main approaches clearly outperformed the majority baseline, showing that they learned useful patterns from the Arabic texts. TF-IDF with Logistic Regression achieved the strongest validation performance across all reported metrics.

Logistic Regression exceeded AraBERT by 3.87 percentage points in Accuracy and 5.26 points in Macro F1. This indicates that the combined word and character TF-IDF representation was highly effective for the available dataset.

The per-class validation results showed that Logistic Regression achieved its highest F1-score for Happiness at 69.77%, followed by Confusion at 55.56% and Optimism at 51.16%. AraBERT achieved its highest validation F1-score for Happiness at 74.42% and Confusion at 56.10%.

The validation predictions are presented visually in:

```
logistic_regression_validation_confusion.png
arabert_validation_confusion.png
```

7.2 Held-Out Test Results

After the preliminary validation comparison, both approaches were evaluated on the 250-record held-out test set.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
TF-IDF + Logistic Regression	46.40%	35.69%	40.60%	**35.81%**	**45.64%**
AraBERT	**49.20%**	**36.00%**	**42.65%**	35.80%	44.63%

AraBERT achieved the higher test Accuracy at 49.20%, compared with 46.40% for Logistic Regression. However, the two approaches obtained almost identical Macro F1-scores: 35.81% for Logistic Regression and 35.80% for AraBERT.

Logistic Regression achieved the higher Weighted F1-score, while AraBERT achieved higher Macro Precision and Macro Recall. Therefore, neither model was better across every test metric.

On the test set, Logistic Regression performed particularly well for Optimism, Happiness, Confusion, and Sadness. Its highest class-level F1-score was 72.29% for Optimism. AraBERT also performed strongly for Confusion and Optimism, achieving F1-scores of 70.73% and 70.59%, respectively.

The test predictions are presented in:

```
logistic_regression_test_confusion.png
arabert_test_confusion.png
```

7.3 Model Comparison

The experimental results show that the preferred model depends on the evaluation stage and metric:

- Logistic Regression achieved the best overall validation performance.
- AraBERT achieved the highest held-out test Accuracy.
- The test Macro F1-scores of both approaches were effectively equal.
- Logistic Regression achieved the higher test Weighted F1-score.
- Both approaches substantially outperformed the majority baseline.

These findings show that the traditional TF-IDF approach remained highly competitive on the limited and noisy Arabic dataset. AraBERT demonstrated stronger test Accuracy, but its

contextual representations did not consistently outperform the combined word and character TF-IDF features across all metrics.

8. Error Analysis

Error analysis was performed using the classification reports, confusion matrices, and exported misclassification files for Logistic Regression and AraBERT. The analysis examined which emotions were difficult to classify and the linguistic or data-related reasons behind incorrect predictions.

The following files were used:

```
logistic_regression_validation_errors.csv
logistic_regression_test_errors.csv
arabert_validation_errors.csv
arabert_test_errors.csv
```

Each file contains the original post, cleaned text, correct emotion label, and predicted label.

8.1 Class Imbalance and Limited Data

The most significant problem was the lack of usable examples for some emotions. Only one usable Neutral example was available in the cleaned training data, and it was retained in the training subset. Consequently, Neutral was absent from validation and both models achieved an F1-score of zero for Neutral on the test set.

The test set also contained only two Surprise examples and four Anticipation examples. Results for these classes are therefore unstable because one correct or incorrect prediction can significantly change their evaluation scores.

8.2 Emotion Overlap

Some emotions share similar vocabulary and may appear together in the same post. Common areas of confusion included:

- Fear and Sadness.
- Happiness and Optimism.
- Confusion and Anticipation.
- Anger and Disgust.
- Pessimism and Sadness.

For example, a post may mention fear-related words while expressing optimism or reassurance. A model focusing on individual words may classify the post as Fear even when its overall meaning is Optimism.

8.3 Negation and Mixed Emotional Cues

Negation was preserved during preprocessing, but it remained difficult for the models to interpret correctly in some cases. A post may contain words associated with fear while explicitly denying fear.

For example:

Text Excerpt	Correct Label	Predicted Label
تخافون من الفايروس... وما يخاف غير اللي متهاون بصلاته	Optimism	Fear
دعواتكم لأمي الله يسعدكم	Sadness	Fear

The first example includes fear-related vocabulary, although the overall message expresses reassurance and optimism. The second includes the positive word `يسعدكم`, but its context indicates sadness and concern.

8.4 Short and Ambiguous Texts

Short posts often do not contain enough context to distinguish between closely related emotions. A brief question may represent Surprise, Confusion, or Anticipation depending on the surrounding conversation.

This issue affected both approaches because the dataset contains many short social-media posts. AraBERT can model surrounding context within a post, but it cannot recover conversational information that is absent from the dataset.

8.5 Dialectal and Informal Arabic

The dataset contains Saudi Arabic, other dialectal expressions, spelling variations, repeated characters, and mixed Arabic-English content. Character-level TF-IDF helps handle some spelling and word-form differences, while AraBERT provides contextual Arabic representations. However, unusual dialectal expressions and informal writing remained difficult to classify.

The original context of many Twitter posts was also unavailable. Therefore, the models had to classify isolated texts without information from replies, previous posts, or conversation threads.

8.6 Logistic Regression Error Patterns

Logistic Regression performed strongly for Happiness, Optimism, Confusion, and Sadness on the test set. However, it struggled with classes having limited training data or overlapping vocabulary.

Its test F1-scores were:

- Optimism: 72.29%.
- Happiness: 70.37%.
- Confusion: 57.14%.
- Sadness: 57.78%.
- Neutral: 0%.
- Surprise: 0%.

The confusion matrix showed that some Fear examples were predicted as Pessimism, Sadness, Anticipation, and Confusion. Disgust examples were frequently predicted as Anger, reflecting the vocabulary overlap between these negative emotions.

8.7 AraBERT Error Patterns

AraBERT achieved strong test performance for:

- Confusion: 70.73% F1.
- Optimism: 70.59% F1.
- Happiness: 64.41% F1.

However, AraBERT achieved zero F1 for Neutral and Surprise. Fear recall was only 11.11%, meaning that it correctly identified a small proportion of the actual Fear examples. Many Fear posts were predicted as Confusion, Pessimism, Anger, or Sadness.

Although AraBERT can model contextual relationships, its performance was limited by the small amount of usable training data and the severe imbalance affecting some emotion classes.

8.8 Main Findings

The main sources of classification errors were:

1. Severe scarcity of usable examples for Neutral and other small classes.
2. Overlapping vocabulary between related emotions.
3. Negation and mixed emotional cues.
4. Short posts with insufficient context.
5. Dialectal expressions and informal spelling.
6. Posts expressing more than one emotion while receiving only one label.
7. Missing conversational context from the original social-media platforms.

Overall, many errors were caused by limitations in the available data rather than by the classification algorithms alone. The results for classes with very small support should therefore be interpreted cautiously.

9. Conclusion

This project developed a supervised multiclass NLP system for detecting emotions in Arabic online text using the ArPanEmo dataset. The work covered the complete classification process, including dataset inspection, Arabic text preprocessing, feature extraction, model development, performance evaluation, and error analysis.

Two main approaches were implemented and compared. The first combined word- and character-level TF-IDF features with Logistic Regression, while the second fine-tuned AraBERT for contextual Arabic emotion classification. A majority-class baseline was also implemented as a simple reference.

On the validation set, Logistic Regression achieved the strongest performance with 50.00% Accuracy and 47.42% Macro F1. On the held-out test set, AraBERT achieved higher Accuracy at 49.20%, compared with 46.40% for Logistic Regression. However, both models achieved nearly identical test Macro F1-scores of approximately 35.8%, while Logistic Regression obtained the higher Weighted F1-score.

These results show that word- and character-level TF-IDF features remain highly competitive for Arabic emotion classification, particularly when the available dataset is relatively small. AraBERT demonstrated the benefit of contextual language representations but did not consistently outperform Logistic Regression across all evaluation metrics.

The main limitation was the lack of available text in the redistributed dataset. After preprocessing, only 1,669 usable records remained, including only one usable Neutral training example. Error analysis also showed that emotion overlap, negation, short context, dialectal writing, and limited class support affected model performance.

Overall, both approaches learned useful emotional patterns and substantially outperformed the majority baseline. The project demonstrates that combining careful Arabic preprocessing with suitable traditional and transformer-based methods can provide meaningful results, while also highlighting the importance of larger and more balanced Arabic emotion datasets.

10. Future Work

Future work should primarily focus on improving data availability and class balance. The original text of legally accessible Twitter posts could be recovered where possible, or the ArPanEmo data could be combined with additional Arabic emotion datasets. More examples are particularly needed for Neutral, Surprise, Anticipation, and Disgust.

Further experiments could optimize the TF-IDF and Logistic Regression approach by testing different n-gram ranges, vocabulary sizes, minimum document frequencies, regularization

values, and class-weighting strategies. Stop-word removal and light Arabic stemming could also be evaluated carefully while preserving negation and emotionally important words.

AraBERT could be improved by tuning the learning rate, batch size, number of epochs, maximum sequence length, and weight decay. Class-weighted loss or data-augmentation techniques could help reduce the effect of class imbalance. Other pretrained Arabic transformer models could also be evaluated and compared.

Repeated stratified validation or cross-validation should be considered when class counts permit, as the current results may be affected by the relatively small validation and test subsets. An ensemble combining Logistic Regression and AraBERT predictions could also be investigated because the two approaches may capture different linguistic patterns.

The project could be extended from single-label to multi-label emotion classification because some posts express more than one emotion. Additional analysis of sarcasm, dialectal expressions, emojis, and conversational context may further improve performance.

Finally, the best-performing model could be integrated into an interactive Arabic-English application that allows users to enter Arabic text and receive the predicted emotion and confidence score. The system could also be evaluated on Jordanian Arabic text to examine how well a model trained mainly on Saudi Arabic generalizes to another dialect.

11. Graphical User Interface (GUI)

A Streamlit GUI was developed to test the trained models interactively. Users can enter Arabic text, select either Logistic Regression or AraBERT, and view the predicted emotion, confidence score, cleaned text, and top predictions.

